SYSTEMATIC SEARCH FOR SINGULARITIES

SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER FLOWS BASED ON THE VOIGT REGULARIZATION

BY

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TITLE: SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER FLOWS BASED ON THE VOIGT REGULARIZATION

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Lay Abstract

A lay abstract of not more 150 words must be included explaining the key goals and contributions of the thesis in lay terms that is accessible to the general public.

The Navier-Stokes equations model the behaviour of viscous incompressible fluids. The question of whether or not the Navier-Stokes equations, or the inviscid Euler equations, are a valid description of fluid mechanics remains unanswered. A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. If the equations form a singularity from smooth initial conditions, then they are not valid descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to not form singularities from smooth initial conditions. However, by decreasing the regularization parameter we approximate Euler flows. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Abstract

Abstract here (no more than 300 words). An abstract of not more than 300 words must be included and indicates the major emphasis of the thesis, new discoveries, and its contribution to knowledge. The style of abstract varies from discipline to discipline but the student should follow an appropriate style. This page must be numbered 'iv'.

The Navier-Stokes equations are a system of partial differential equations used to model the behaviour of viscous incompressible fluids. The question of whether or not unique classical solutions of the Navier-Stokes equations, and inviscid Euler equations, exist for all time, given smooth initial conditions, remains unanswered for the three-dimensional problem.

A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. The formation of a singularity from smooth initial conditions would invalidate the equations as accurate descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to be globally well-posed from smooth initial conditions. However, by decreasing the regularization parameter the solutions of the Euler-Voigt equations converge to those of the Euler equations. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Your Dedication

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Acknowledgements

Acknowledgements go here. An expression of thanks for assistance given by the Supervisor of research and by others should be either set forth on a separate page or incorporated into the Preface (if there is one). These and all subsequent preliminary pages listed under (f), (g) and (h) must be numbered in lower case Roman numerals, i.e., 'iv', 'v', 'vi' etc.

Dr. Bartosz Protas, Dr. Xinyu Zhao, Dr. Jörn Davidsen, members of committee, Parents

I would like to thank Dr. Bartosz Protas for his support and guidance on my research project. I also want to thank Dr. Jörn Davidsen of the Complexity Science Group for supervising my undergraduate research and inspiring me to pursue computational physics.

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Notation, and Abbreviations

Notation

x	Plain-text to denote scalar x
$\boldsymbol{x}$	Bold-text to denote vector x
$\boldsymbol{x}^{\intercal}$	Transpose of vector x
I	Identity Operator
$\boldsymbol{\nabla}$	Gradient Operator
Δ	Laplacian Operator
$:=$	Equal to by definition

Abbreviations

PDEs	Partial Differential Equations
1D	one-dimensional or one dimension
2D	two-dimensional or two dimensions

3D three-dimensional or three dimensions

TGV Taylor Green Vortex

Declaration of Academic Achievement

Dr. Zhao developed most of the mathematics for the subspace V (2.26), and Theorem 2.1.3 and its proof. Dr. Zhao also developed the mathematics for much of the Euler-Voigt optimization problem described in Chapter 3, including the objective functional (3.32), the linearization of the Euler-Voigt equations (3.39), the adjoint system (3.40), and the gradient of the objective functional (3.45). The Riemannian conjugate gradient method for the Euler-Voigt optimization problem in Section 3.3 was developed by Dr. Zhao and Dr. Protas. This includes the tangent space (3.46), the projection operator (3.49), the search direction (3.53), and step size (3.55).

The retraction operator (3.50), and the vector transport operation (3.51) were originally developed by Dr. Zhao and Dr. Protas but I modified them to suit the present problem. The formulation of the manifold $\mathcal{M}_C$ (2.28) was based on work by Dr. Zhao, but the calculation of the constraint C was my work. This included the calculations of the $\dot{H}^1$ seminorm (2.30) and vorticity (2.31) of the general Taylor Green Vortex, which are included in Appendices F and G. The results in Appendices A, B, C, and D are my re-derivations of Dr. Zhao's original results. The derivation in Appendices G and H are my re-derivations of the well known Euler and Euler-Voigt scaling properties.

For the numerical models, Dr. Zhao created the original code base which I modified to suit this project. The determination of the numerical model constants, the time and spatial resolutions, regularization parameter α, and Gevrey parameter σ were also my work. These values were validated through analysis of the stability and accuracy of the model, conservation of the α-energy, comparison of the effect of different values of σ, examination of the solutions' Fourier spectrum, and the kappa test, all of which were my work and are described in Chapter 4. I also validated the results of the numerical models through comparison with the relevant findings of [3] and [17]. I performed all of the computations to obtain the final numerical results. The findings and conclusions in Chapters 5 and 6 are my work. The Python code used to perform the data analysis and produce the graphs is also my own work.

1 Introduction

1.1 Singularity Formation in Models of 3D Fluid Flows

The three-dimensional (3D) Navier-Stokes equations are a set of partial differential equations (PDEs) that describe the motion of homogeneous, viscous, incompressible fluids. The Navier-Stokes equations on the domain $\Omega \subseteq \mathbb{R}^3$ are defined as

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p - \nu \Delta \boldsymbol{u} = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.1a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.1b}$$

$$\boldsymbol{u}(x, 0) = \boldsymbol{\eta} \qquad \text{in } \Omega \tag{1.1c}$$

where $\boldsymbol{u} = [u_1, u_2, u_3]^\intercal$ is the fluid velocity field, p is the scalar pressure, $\nu > 0$ is the coefficient of kinematic viscosity, and T is the length of the time interval. The field $\boldsymbol{\eta}$ is the initial condition assumed to be divergence-free. Finally, the gradient operator $\boldsymbol{\nabla} = [\partial_1, \partial_2, \partial_3]^\intercal$ operator is used to define the gradient of the pressure, the tensor $[\boldsymbol{\nabla}\boldsymbol{u}]_{ij} = \partial_j u_i$ for $i, j = 1, 2, 3$, the divergence of the velocity field, and the Laplacian operator $\Delta \boldsymbol{u} = \boldsymbol{\nabla} \dot{c}(\boldsymbol{\nabla}\boldsymbol{u})$. Because of their generality, the Navier-Stokes equations are utilized to model the behaviour of systems in fields as disparate as aero/hydrodynamics, weather prediction, biophysics, and astrophysics [8][10]. Despite the numerous fields that make use of the 3D Navier-Stokes equations, and extensive research examining their properties, it is only known that weak Leray-Hopf solutions exist globally in time [19]. However, these solutions may contain singularities and their uniqueness remains unproven in most cases. It also yet to be proven whether or not smooth classical solutions defined in the pointwise sense exist globally in time [25]. The space of smooth functions and all other function spaces utilized in this analysis are discussed in more detail in Appendix A. In fact, this is the basis of one of the seven millennium problems posed by the Clay Mathematics Institute which offers a substantial monetary prize for the confirmation of the existence, or non-existence, of a classical solution corresponding to smooth initial conditions [11]. It is common for scientific research that focuses the fundamental properties of the Navier-Stokes equations to utilize either the unbounded domain $\Omega := \mathbb{R}^3$, the axisymmetric domain, or the periodic 3D torus $\Omega := \mathbb{T}^3_L$ with domain size $L > 0$ [25]. For this analysis, we use the 3D torus of unit size $(L = 1)$ with periodic boundary conditions.

A central question in fluid mechanics is whether or not the Navier-Stokes equations are globally well-posed or if singularities can form from smooth initial conditions. A singularity occurs when the derivative of the velocity field becomes infinite at a single point [22]. The formation of a singularity in a fluid flow, from smooth initial conditions, would invalidate the Navier-Stokes equations as an accurate description of fluid flows [25]. The question about singularity formation from smooth initial conditions also remains open for the Euler equations [13]. The Euler equations are the inviscid form of the Navier-Stokes equations, obtained by setting $\nu = 0$

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.2a}$$

$$\boldsymbol{\nabla} \cdot \mathbf{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.2b}$$

$$\boldsymbol{u}(x, 0) = \boldsymbol{\eta} \qquad \text{in } \Omega. \tag{1.2c}$$

The Euler equations are often studied because they are simpler to analyse than the Navier-Stokes equations due to the absence of the viscous dissipation. The local existence of classical solutions to the Euler equations was proven in [16] for solutions defined in Sobolev spaces.

Theorem 1.1.1 ([16]). *If $\boldsymbol{\eta} \in H^m(\mathbb{T}^3)$ for some $m > 5/2$ and satisfies $\boldsymbol{\nabla} \cdot \boldsymbol{\eta} = 0$, then there exists a time $T = T(\|\boldsymbol{\eta}\|_{H^m}) > 0$ such that (1.2) has a unique solution $\boldsymbol{u}(\cdot; \boldsymbol{\eta}) \in C([0, T]; H^m) \bigcap C^1([0, T]; H^{m-1})$.*

If a singularity formation is to occur in the solutions to the Euler equations, it must form after the minimum local existence time defined in Theorem 1.1.1. There are many different conditional regularity results for determining if a singularity has formed, the best known of which is the Beale-Kato-Majda (BKM) criterion [1] which states

Theorem 1.1.2 ([1]). *A smooth solution $\boldsymbol{u}$ of the Euler system develops a singularity at $t = t_0$ if and only if*

$$\lim_{t \to t_0} \int_0^t \|\boldsymbol{\omega}\|_{L^\infty} d\tau = \infty, \qquad \boldsymbol{\omega} = \nabla \times \boldsymbol{u}, \tag{1.3}$$

where $\|\cdot\|_{L^\infty}$ is the standard Lebesgue L^∞ norm.

A key difference between the Navier-Stokes and Euler equations is that there is more numerical evidence that indicates singularity formation in Euler is possible from smooth initial conditions. For example, [2] and [3] presented numerical evidence for singularity formation in the Euler equations on a 3D torus ($L = 2\pi$). Hou et al. produced several papers which provide numerical evidence for singularity formation in the 3D axisymmetric Euler equations [21][15][14] and which proved singularity formation in the 3D axisymmetric Euler equations with rigid boundaries and periodic boundary conditions in the axial direction [6][7]. We make special note of the optimization-based approach of [27], which utilized a Riemannian conjugate gradient method to identify 'extreme' initial conditions, and offers numerical results consistent with singularity formation in Euler flows on the 3D unit torus. Finally, [17] which provides numerical evidence for singularity formation in the Euler equations based on the Voigt regularization. These final two papers are of special significance for us because they form the primary basis for the motivation and methodology of this analysis.

1.2 3D Euler-Voigt Equations

The Euler-Voigt equations are an inviscid regularization of the Euler equations (1.2) which have been shown to be globally well-posed and so do not form singularities from smooth initial conditions [5]. The Euler-Voigt equations are defined as

$$-\alpha^2 \partial_t \Delta \boldsymbol{u}_\alpha + \partial_t \boldsymbol{u}_\alpha + \boldsymbol{u}_\alpha \cdot \boldsymbol{\nabla} \boldsymbol{u}_\alpha + \boldsymbol{\nabla} p_\alpha = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.4a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u}_\alpha = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.4b}$$

$$\boldsymbol{u}_\alpha(x, 0) = \boldsymbol{\eta}_\alpha \qquad \text{in } \Omega, \tag{1.4c}$$

where the regularization is accomplished with the addition of the term $-\alpha^2\partial_t\Delta\boldsymbol{u}_\alpha$ in which $\alpha > 0$ is the regularization parameter with units of length. This term stabilizes the system without dissipation, which would remove energy from the system [17]. The regularization term acts as a spectral filter and suppresses the development of small scale structures with length-scale smaller than α [23]. We introduce the subscript α to distinguish between solutions of (1.2) and (1.4) i.e., $\boldsymbol{u}$ and $\boldsymbol{u}_\alpha$.

For (1.2) the total kinetic energy is conserved so long as the solution remains well-posed in the classical sense

$$E_k(t) := \frac{1}{2}\|\boldsymbol{u}\|_{L^2}^2 = E_k(0), \tag{1.5}$$

where the norm $\|\cdot\|_{L^2}$ here is the standard Lebesgue L^2 norm. For (1.4) the conserved quantity is known to be the α-energy. The α-energy is defined as

$$E_\alpha(t) := \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0). \tag{1.6}$$

1.3 Outline of the Thesis

The goal of this analysis is to search for possible singularities in the Euler flows corresponding to smooth initial conditions. This search will be performed based on a PDE-constrained optimization problem in which we will identify initial condition that produce the most 'extreme' behaviour in Euler-Voigt flows. In Chapter 2 we present the properties of the Euler-Voigt equations, including the blow-up criteria, scaling property, and manifold over which we search for initial conditions. In Chapter 3 we present the formulation of the optimization problem. This includes the description of our objective functional, the gradient calculation, and the Riemannian conjugate gradient method used to compute our search direction and iterative relation. In Chapter 4 we describe the numerical methods used to perform the calculations, including approximation in space and time, Gevrey parameters, and the regularization parameter. In addition, we validate the computations using the Fourier spectrum, kappa test, and conservation of $\alpha-$energy. In Section 5 we present our results. We begin with the solutions of the Euler-Voigt equations corresponding to the Taylor Green Vortex without optimization as $\alpha \to 0$. These results indicate that singularity formation does not occur within the time window. We then present our optimization results as $\alpha \to 0$, which indicate that RESULTS RESULTS RESULTS. Finally, in Chapter 6 we summarize our findings and present our conclusion.

2 Properties of the Euler-Voigt Equations

2.1 Finite Time Blow-up Criterion

In order to begin our search for finite-time singularity formation, we must first describe the criteria used to detect singularity formation in Euler flows. In [18] a new criterion for identifying singularity formation in Euler flows was developed which utilizes the Euler-Voigt equations. This criterion relies on the convergence of the solutions of (1.4) to the solutions of (1.2) as $\alpha \to 0$.

Theorem 2.1.1 ([18])**.** *Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$, let $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\Omega)) \cap C^1([0,T], H^{s-1}(\Omega))$ be the corresponding solutions to (1.2) and (1.4), respectively, where $0 < T < T_{max}$ and T_{max} is the maximal time for which a solution to (1.2) exists and is unique. For all $t \in [0,T]$, we have*

$$\begin{aligned} \|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq \\ \left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2 (e^{Ct} - 1). \end{aligned} \tag{2.7}$$

Based on this theorem we know that if

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \text{as } \alpha \to 0, \tag{2.8}$$

then for all $t \in [0, T]$,

$$\|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \to 0. \tag{2.9}$$

Therefore, solutions of the Euler-Voigt system (1.4) converge to the corresponding Euler (1.2) flows in H^1 uniformly in time as $\alpha \to 0$. Using this result, the following theorem was proved in [17].

Theorem 2.1.2. *Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$, with $\boldsymbol{\nabla} \cdot \eta_\alpha = 0$ and let $\boldsymbol{u}_\alpha$ be the corresponding unique solution of (1.4). Suppose that there exists a time $T^* > 0$ such that*

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0. \tag{2.10}$$

Then, the corresponding solution of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^*]$.

We note that this theorem relies on the solutions of (1.2) and (1.4) starting from the same initial condition, i.e., $\boldsymbol{\eta} = \boldsymbol{\eta}_\alpha$. This necessarily satisfies (2.8) as $\alpha \to 0$. However, in order to use this result in our study, we need to generalize Theorem 2.1.2 for the case (2.8) where $\boldsymbol{\eta} \neq \boldsymbol{\eta}_\alpha$. Therefore, we modify Theorem 5.2 from [18] as follows.

Theorem 2.1.3 (X. Zhao, private communication)**.** *Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$ satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \textit{as } \alpha \to 0, \tag{2.11}$$

let $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$ be the corresponding solutions to

(1.2) and (1.4), respectively. If there exists a finite time $T^* > 0$ *such that the solutions* $\boldsymbol{u}_\alpha$ *satisfy*

$$\limsup_{\alpha\to 0}\left(\alpha^2 \sup_{t\in[0,T^*]} \|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) > 0, \tag{2.12}$$

then $\boldsymbol{u}$ *develops a singularity in the interval* $[0, T^*]$.

Proof. Suppose that the solution $\boldsymbol{u}$ of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have, cf. (1.6)

$$\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}\boldsymbol{\eta}_\alpha\|_{L^2}^2, \tag{2.13}$$

thus

$$\alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}\boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{2.14}$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{2.15}$$

Taking the limsup as $\alpha \to 0$, we obtain

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]} \alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) = \limsup_{\alpha\to 0}\left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right). \tag{2.16}$$

We now make use of Theorem 2.1.1, which demonstrates that the convergence of the solutions of the Euler-Voigt equations (1.4) to the solution of the Euler equations

(1.2) is uniform on $[0, T^*]$. Therefore we can reduce the above equation to

$$\limsup_{\alpha \to 0} \left(\sup_{t \in [0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) = \|\boldsymbol{\eta}\|_{L^2}^2 - \|\boldsymbol{u}(t)\|_{L^2}^2. \tag{2.17}$$

Since the energy of regular solutions of the Euler equations is conserved, we have

$$\limsup_{\alpha \to 0} \left(\sup_{t \in [0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) = 0, \tag{2.18}$$

which contradicts the hypothesis in the theorem that the above expression is positive. Equality (2.17) holds because

$$\begin{aligned}
\limsup_{\alpha \to 0} \left| \inf_{t \in [0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2} - \|\boldsymbol{\eta}\|_{L^2} \right| &\leq \limsup_{\alpha \to 0} \|\boldsymbol{u}_\alpha(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
&\leq \limsup_{\alpha \to 0} \|\boldsymbol{u}_\alpha(t_\alpha) - \boldsymbol{u}(t_\alpha)\|_{L^2}^2 + \|\boldsymbol{u}(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
&= 0,
\end{aligned}$$

where the reverse triangle and triangle inequalities were used and t_α is the time at which

$$\inf_{t \in [0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2} = \|\boldsymbol{u}_\alpha(t_\alpha)\|_{L^2}. \tag{2.20}$$

□

2.2 Initial Conditions

We have established Theorem 2.1.3 which we will use to identify singularity formation in Euler flows corresponding to smooth initial conditions. In order to formulate our optimization problem we must now describe the function space we perform our search in. The Theorems (2.1.1), (2.1.2), and (2.1.3) presented in Section (2.1) require initial conditions $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$. However, the pseudo-spectral methods used in this study to numerically solve system (1.4) require the functions to be analytic in order to have exponential convergence [4]. For the Navier-Stokes equations (1.1), the viscous term acts to instantaneously regularize the solution in space. Therefore, if the initial condition has only Sobolev regularity, the resulting flow becomes real-analytic (with Gevrey regularity defined below) at $t = 0^+$. This regularity allows the solution to achieve exponential convergence.

The Euler equations (1.2) do not include any viscous effects and so the solution simply stays at the same level of smoothness as the initial condition, unless blow-up occurs. For Euler-Voigt the regularization term keeps the solution globally well-posed but does not improve the regularity [18]. Therefore, we must have analytic initial conditions to make use of the pseudo-spectral methods. For all the models considered here, a requirement is that the flow is divergence free at all times, including the initial conditions. Finally, on a periodic domain the quantity $\int_\Omega \boldsymbol{\eta}_\alpha \, d\boldsymbol{x}$ is an invariant of motion for solutions of the Euler-Voigt equations. Due to this, we assume without loss of generality that the initial conditions have a mean of zero. In order to identify initial conditions that satisfy these criteria, we first consider analytic initial conditions that belong to the Gevrey class

$$G^\sigma := \left\{ \boldsymbol{v} \in H^s(\mathbb{T}^3) : \|\boldsymbol{v}\|_{G^\sigma} = \langle \boldsymbol{v}, \boldsymbol{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0, \tag{2.21}$$

with the norm defined as

$$\forall \boldsymbol{v}, \boldsymbol{u} \in G^{\sigma}, \quad \langle \boldsymbol{v}, \boldsymbol{u} \rangle_{G^{\sigma}} := \sum_{\boldsymbol{j} \in \mathbb{Z}^3} (1 + |2\pi \boldsymbol{j}|^2)^s e^{4\pi\sigma|\boldsymbol{j}|} \widehat{\boldsymbol{v}}_{\boldsymbol{j}} \cdot \overline{\widehat{\boldsymbol{u}}_{\boldsymbol{j}}} \tag{2.22a}$$

$$= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} \boldsymbol{v} \cdot \boldsymbol{u} \, d\boldsymbol{x}, \tag{2.22b}$$

where hat denotes the Fourier coefficient

$$\widehat{\boldsymbol{u}}_k = \int_{\Omega} e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \boldsymbol{u}(\boldsymbol{x}) d\boldsymbol{x}, \tag{2.23}$$

overbar denotes complex conjugation, and the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D|\boldsymbol{v}}\right]_{\boldsymbol{j}} := 2\pi|\boldsymbol{j}|\hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|}\boldsymbol{v}}\right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|}\hat{\boldsymbol{v}}_{\boldsymbol{j}}. \tag{2.24}$$

The parameter σ determines the relative 'size' of the Gevrey space

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s, \quad 0 < \sigma_1 < \sigma_2, \tag{2.25}$$

which is proved in Appendix E. The value of σ is chosen in Section 4.3. In order to satisfy the divergence free condition, we introduce the following subspace in which optimal initial conditions will be sought

$$V := \{\boldsymbol{v} \in G^{\sigma} : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \quad \int_{\mathbb{T}^3} \boldsymbol{v} \, d\boldsymbol{x} = 0\}. \tag{2.26}$$

2.3 Scaling Symmetries of Euler and Euler-Voigt Systems

Both (1.2) and (1.4) possess a scaling property. Given a solution $\boldsymbol{u}$, or $\boldsymbol{u}_\alpha$, we can construct a new solution

$$\boldsymbol{v}(\boldsymbol{y},\tau) = \frac{1}{\gamma}\boldsymbol{u}(\beta\boldsymbol{x},\lambda t), \qquad \boldsymbol{y} = \beta\boldsymbol{x}, \qquad \tau = \lambda t \tag{2.27}$$

where $\lambda, \gamma, \beta > 0$ satisfy $\beta/(\lambda\gamma) = 1$, which is proven in Appendices H and I, respectively for the Euler (1.2) and Euler-Voigt (1.4) equations. Our goal is to find a time interval $[0, T]$ that is sufficiently large so that it is outside the interval of local existence established by Kato's theorem 1.1.1 and so allows for the potential of singularity formation in Euler flows. Due to this scaling property of the Euler-Voigt equations (2.27), the blow-up time of the solution will depend on the 'size' of our solution (γ) and the spatial scale (β). Our domain size will not change, and so in order to keep consistent time scale we must fix γ between all initial conditions. To accomplish this, we restrict our discussion to initial conditions with a fixed $\dot{H}^1$ seminorm which belong to a closed manifold $\mathcal{M}_C \subset V$ defined as cf. (2.26)

$$\mathcal{M}_C = \{\boldsymbol{v} \in V : \|\boldsymbol{v}\|_{\dot{H}^1} = C\}, \tag{2.28}$$

where $C > 0$ is a constant. We construct the manifold by fixing the $\dot{H}^1$ seminorm because of the form of our objective functional used in the optimization search, which will be described in Chapter 3. The question we must answer now is what value of C we use. To determine this we look to our first initial condition, the 3D Taylor Green

Vortex (TGV). The TGV in the domain $\Omega = [0, L]^3$ has the form

$$\boldsymbol{\eta}_{TGV}(x) = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0 \end{bmatrix}, \tag{2.29}$$

for some $V_0 \neq 0$. The $\dot{H}^1$-seminorm of (2.29) is

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3L}\pi V_0, \tag{2.30}$$

which is proven in Appendix F. In addition, the BKM criterion (1.1.2) uses the quantity $\|\boldsymbol{\omega}\|_{L^\infty}$ to determine singularity formation. For the TGV

$$\|\boldsymbol{\omega}\|_{L^\infty} = \|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 4\pi V_0/L, \tag{2.31}$$

which is proven in Appendix G. In [3], the authors provided numerical evidence that blow-up of (1.2) may occur around $T^* \approx 4$ for initial condition (2.29) with parameters $L = 1$ and $V_0 = 1$. This result provides the baseline for our study and so we must match this time window so that singularity formation identified by (2.1.3), if it happens, occurs at approximately the same time as in [3]. Therefore, we set $\lambda = 1$. Our Euler-Voigt system is defined on the domain $\Omega = [0, 1]^3$ and so we set $\beta = 1/2\pi$. Therefore, from (2.27) we set $\gamma = 1/2\pi$. The model we are aiming to match uses $V_0 = 1$ and so to match the time window we use $V_0 = 1/2\pi$.

These considerations give $\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$ and $\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 2$ for the initial condition (2.29). Therefore, we can restrict our discussion of the TGV to initial conditions with $\dot{H}^1$ seminorm which belong to a closed manifold (2.28) with $C = \sqrt{3}/2$. In order to select initial conditions within this manifold, we will first select initial conditions $\boldsymbol{\eta} \in V$ and then apply a retraction function to ensure it has $\|\cdot\|_{\dot{H}^1} = \sqrt{3}/2$. To verify that this selection of the $\dot{H}^1$ seminorm of the initial condition is correct, we compare this scaling to the results shown in Figure 1b of [3]. This figure shows the evolution of the L^∞ norm of the vorticity over time.

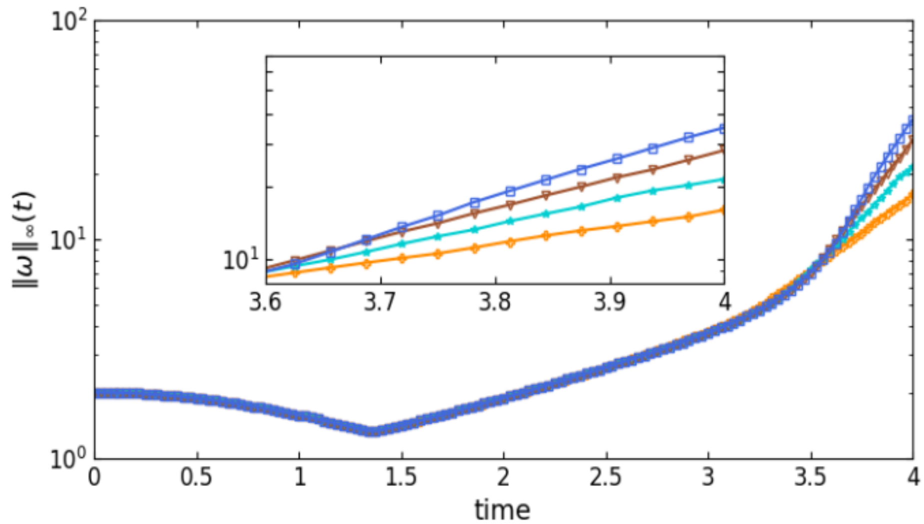

Figure 2.1: Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

We present our reproduction in Figure 2.1 which shows that the evolution of the vorticity at each resolution in our scaled model matches exactly to the corresponding computations shown in Figure 1b in [3]. Therefore, our construction of the manifold $\mathcal{M}_C$ is correct.

3 Statement of the Optimization Problem

3.1 Formulation of the Problem

With our initial conditions and blowup criteria defined, we now construct the PDE-constrained optimization problem. We begin with the objection functional $\Phi_{\alpha,T} : V \to \mathbb{R}^+$, which is defined as

$$\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{L^2}^2, \tag{3.32a}$$

$$:= \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2, \tag{3.32b}$$

where $\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)$ is the solution of the Euler-Voigt equations (1.4) obtained at time T with the initial condition $\boldsymbol{\eta}_\alpha \in V$ and regularization parameter α. Given $\alpha, T \in \mathbb{R}^+$, the optimization problem will find the initial condition that maximizes the objective functional

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta}_\alpha \in M_C}{\operatorname{argmax}} \, \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha). \tag{3.33}$$

The objective functional is maximized using an Riemannian conjugate gradient iterative relation described in Section 3.3. This method is utilized because it accelerates the gradient ascent method which is necessary given the computational cost of each iteration. The solution to the problem is the value of $\boldsymbol{\eta}^n_{\alpha,T}$ as the iteration number $n \to \infty$. The same method was used in [27] to construct a PDE-constrained optimization problem for the Euler equations to search for initial conditions $\boldsymbol{\eta}$ that produce 'extreme' flows.

Our goal is to first fix T and solve the optimization problem for a sequence of decrease values of α that converge to 0. We examine the results for a time interval before the expected point singularity formation should occur and again for a time interval after. From Theorem 2.1.3 if

$$\|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T}\|^2_{L^2} + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T})\|^2_{L^2} \to 0, \quad \text{as } \alpha \to 0, \tag{3.34}$$

and

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0,T^*]} \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) \right) > 0, \tag{3.35}$$

then $\boldsymbol{u}$ develops a singularity in the interval $[0, T^*]$ from initial condition $\boldsymbol{\eta}$. From the results of [3], the expected time of singularity formation in Euler is $T^* \approx 4$. Therefore, we perform the optimization problem for $T = 3$, where blow-up should not occur, and $T = 5$ where it may occur.

3.2 Gradient of Objective Function

The optimization problem is solved using an iterative relation which uses an optimal search direction and distance to create the next iteration's initial condition. We compute the optimal search direction using the gradient of the objection functional $\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)$. This value is computed using an adjoint method which depends on solving a properly defined adjoint system backwards in time. To derive the gradient, we first introduce the Gâteaux (directional) differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') : V \times V \to \mathbb{R}$. The first order finite difference equation for the Gateaux differential is,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}\left[\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha+\epsilon\boldsymbol{\eta}'_\alpha)-\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)\right]. \tag{3.36}$$

This represents the change in $\Phi_{\alpha,T}$ from a small perturbation proportional to ϵ in the direction $\boldsymbol{\eta}'_\alpha$ from the initial condition $\boldsymbol{\eta}_\alpha$. By fixing the initial condition $\boldsymbol{\eta}_\alpha$, we can view the Gâteaux differential as a bounded linear functional on V. Therefore, it can also be represented as an inner product on the Gevrey space using the Riesz representation theorem [27] [20]

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}'_\alpha) = \langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha),\boldsymbol{\eta}'_\alpha\rangle_{G^\sigma}. \tag{3.37}$$

Substituting the second definition of the objective functional (3.32) into the finite difference form of the Gâteaux differential (3.36), we derive the Gâteaux differential in the form of the $\dot{H}^1$ semi-inner product and the L^2 inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T),\boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} \tag{3.38a}$$

$$= \left\langle \boldsymbol{u}^*_\alpha(\cdot,0),\boldsymbol{u}'_\alpha(\cdot,0)\right\rangle_{L^2} \tag{3.38b}$$

The derivations for both forms are shown in Appendix B. In (3.38) the value $\boldsymbol{u}'_\alpha$ is the solution of the linearization of the Euler-Voigt system (1.4) around its solution $\boldsymbol{u}_\alpha$. The linearization of the Euler-Voigt system is defined as

$$\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix} := \begin{bmatrix}\partial_t \boldsymbol{u}'_\alpha + (I-\alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p') = 0\\ \nabla\cdot\boldsymbol{u}'_\alpha = 0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \tag{3.39a}$$

$$\boldsymbol{u}'_\alpha(x,0) = \boldsymbol{\eta}'_\alpha \tag{3.39b}$$

where $\boldsymbol{\eta}'_\alpha$ and p' are the perturbations of the initial condition and pressure, respectively. The derivation of the linearized form of the Euler-Voigt equations is shown in Appendix C. The value $\boldsymbol{u}^*_\alpha$ in (3.38) is the solution of the adjoint system

$$\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*_\alpha\end{bmatrix} := \begin{bmatrix}-\partial_t \boldsymbol{u}^*_\alpha - (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha + (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha)^\intercal)\boldsymbol{u}_\alpha - \boldsymbol{\nabla}p^*_\alpha\\ -\boldsymbol{\nabla}\cdot\boldsymbol{u}^*_\alpha\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \tag{3.40a}$$

$$\boldsymbol{u}^*_\alpha(\boldsymbol{x},T) = 2|D|^2\boldsymbol{u}_\alpha(\boldsymbol{x},T) \tag{3.40b}$$

which is derived in Appendix D by performing integration by parts, in both space and time, on the following duality-pairing relation

$$\begin{aligned}
\left(\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) &:= \int_0^T\int_{\mathbb{T}^3}\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\,d\boldsymbol{x}\,dt\\
&= \left(\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) + \int_{\mathbb{T}^3}\boldsymbol{u}'_\alpha(\boldsymbol{x},T)\cdot\boldsymbol{u}^*_\alpha(\boldsymbol{x},T)\,d\boldsymbol{x}\\
&\quad - \int_{\mathbb{T}^3}\boldsymbol{\eta}'_\alpha(\boldsymbol{x})\cdot\boldsymbol{u}^*_\alpha(\boldsymbol{x},0)\,d\boldsymbol{x}\\
&= 0.
\end{aligned} \tag{3.41}$$

Using the adjoint system we can now define the gradient. We take the Gâteaux differential in the form of the L^2 inner product (3.38) and in the Riesz representation form (3.37)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') = \langle \boldsymbol{u}^*_\alpha(\cdot,0), \boldsymbol{u}'_\alpha(\cdot,0)\rangle_{L^2} = \langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha\rangle_{G^\sigma}\,. \tag{3.42}$$

Using the definition of the Gevrey inner product (**??**), the L^2 inner product can be converted to the Gevrey inner product,

$$\langle \boldsymbol{u}^*_\alpha(\cdot,0), \boldsymbol{u}'_\alpha(\cdot,0)\rangle_{L^2} = \left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'_\alpha(x)\right\rangle_{G^\sigma} \tag{3.43}$$

Therefore, we have two forms of the Gâteaux differential as a Gevrey inner product

$$\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha\rangle_{G^\sigma} = \left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'_\alpha(x)\right\rangle_{G^\sigma}\,. \tag{3.44}$$

Therefore, the gradient of the objective functional is

$$\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1+|D|^2)^{-3} e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0) \tag{3.45}$$

The adjoint system (3.40) is a terminal value problem. Therefore, to compute the gradient we first evolve the Euler-Voigt system forward from $\boldsymbol{\eta}_\alpha$ to compute $\boldsymbol{u}_\alpha(\boldsymbol{x},T)$, from which we obtain the terminal value of (3.40). Finally, we evolve the adjoint system backwards in time to obtain the initial value $\boldsymbol{u}^*_\alpha(\boldsymbol{x},0)$ which is used to compute the gradient (3.45).

3.3 Riemannian Conjugate Gradient Method

The optimization solution looks for an initial condition $\boldsymbol{\eta}_\alpha \in M_C$ that maximizes our objective functional. Therefore, not only must our starting initial condition $\boldsymbol{\eta}^0_{\alpha,T}$ have a $\dot{H}^1$-seminorm of $\sqrt{3}/2$, but so too must each following iteration $\boldsymbol{\eta}^n_{\alpha,T}$. We ensure this constraint is upheld using a Riemannian conjugate gradient method modified from the method used in [25] and [9]. This method has three steps:

1. Project the gradient $\nabla\Phi_T(\boldsymbol{\eta}_\alpha)$ onto the tangent space to $\mathcal{M}_C$ at $\boldsymbol{\eta}_\alpha$.
2. Perform a suitable vector transport operation in order to construct a Riemannian conjugate ascent direction using the previous search direction.
3. Retract the resulting state back to the constraint manifold $\mathcal{M}_c$.

We first define the tangent space to $\mathcal{M}_C$ as

$$T_{\boldsymbol{v}}\mathcal{M}_C = \left\{\boldsymbol{z} \in V : \langle \boldsymbol{v}, \boldsymbol{z}\rangle_{\dot{H}^1} = 0\right\}. \tag{3.46}$$

For any $\boldsymbol{z} \in \mathcal{T}_{\boldsymbol{v}}\mathcal{M}_C$, this condition can be expressed as

$$\begin{aligned} \langle \boldsymbol{v}, \boldsymbol{z}\rangle_{\dot{H}^1} &= \int_{\mathbb{T}^3} |D|\boldsymbol{v} \cdot |D|\boldsymbol{z} d\boldsymbol{x} & \text{(3.47a)}\\ &= \int_{\mathbb{T}^3} (1+|D|^2)^3 e^{2\sigma|D|}\left((1+|D|^2)^{-3}e^{-2\sigma|D|}|D|^2\boldsymbol{v}\right)\cdot \boldsymbol{z}\, d\boldsymbol{x} & \text{(3.47b)}\\ &= \left\langle (1+|D|^2)^{-3}e^{-2\sigma|D|}|D|^2\boldsymbol{v}, \boldsymbol{z}\right\rangle_{G^\sigma} = 0. & \text{(3.47c)} \end{aligned}$$

Thus, at each point $\boldsymbol{v} \in M_C$, the tangent space $\mathcal{T}_{\boldsymbol{v}}\mathcal{M}_C$ can be characterized using the unit 'normal' vector given by

$$\boldsymbol{n}_{\boldsymbol{v}} = \frac{(1+|D|^2)^{-3}e^{-2\sigma|D|}|D|^2\boldsymbol{v}}{\|(1+|D|^2)^{-3}e^{-2\sigma|D|}|D|^2\boldsymbol{v}\|_{G^\sigma}} \tag{3.48}$$

which allows us to construct the projection operator $\mathcal{P}_{\mathcal{T}_{\boldsymbol{v}}M_C} : V \to \mathcal{T}_{\boldsymbol{v}}\mathcal{M}_C$,

$$\forall \boldsymbol{w} \in V, \quad \mathcal{P}_{\mathcal{T}_{\boldsymbol{v}}\mathcal{M}_C}\boldsymbol{w} = \boldsymbol{w} - \langle \boldsymbol{n}_{\boldsymbol{v}}, \boldsymbol{w} \rangle_{G^\sigma} \boldsymbol{n}_{\boldsymbol{v}}. \tag{3.49}$$

As stated previously, the initial conditions at each iteration must have a $\dot{H}^1$-seminorm of $\sqrt{3}/2$. In order to ensure this we introduce the Retraction operator

$$\mathbf{R}(\boldsymbol{v}) = \frac{\sqrt{3}}{2}\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_{\dot{H}^1}}, \qquad \boldsymbol{v} \neq \mathbf{0}. \tag{3.50}$$

The Reimannian method uses the search direction of the n-th iteration to compute the search direction of the $(n+1)$-th iteration. This is done by a linear combination of the current direction and a 'momentum' term based on the previous iteration's search direction. However, the two search directions belong to different tangent spaces to $\mathcal{M}_C$ and thus cannot be directly added. In order to allow algebraic operations to be performed on the vectors of the two tangent spaces we introduce the vector transport operation

$$\mathcal{T}\mathcal{M}_C \oplus \mathcal{T}\mathcal{M}_C \to \mathcal{T}\mathcal{M}_C : (\boldsymbol{\varphi}, \boldsymbol{\xi}) \mapsto \Gamma_{\boldsymbol{\varphi}}(\boldsymbol{\xi}) \in \mathcal{T}\mathcal{M}_C,$$

where $\mathcal{T}\mathcal{M}_C = \bigcup_{\boldsymbol{v}\in\mathcal{M}_C} \mathcal{T}_v\mathcal{M}_C$ is the tangent bundle on $\mathcal{M}_C$. The operation transports the vector field $\boldsymbol{\xi}$ along the manifold M_C in the direction of $\boldsymbol{\varphi}$. This allows us to map search directions between the tangent spaces of two consecutive iterations.

The operator is defined via differentiated retraction

$$\begin{aligned}\Gamma_{\varphi}(\xi) &:= \frac{d}{dt}\mathbf{R}(\varphi + t\xi)\big|_{t=0}\\ &:= \frac{\sqrt{3}}{2}\frac{1}{\|\boldsymbol{u}+\varphi_{\boldsymbol{u}}\|_{\dot{H}^1}}\left[\xi_{\boldsymbol{u}} - \frac{\langle \boldsymbol{u}+\varphi_{\boldsymbol{u}}, \xi_{\boldsymbol{u}}\rangle_{\dot{H}^1}}{\|\boldsymbol{u}+\varphi_{\boldsymbol{u}}\|^2_{\dot{H}^1}}(\boldsymbol{u}+\varphi_{\boldsymbol{u}})\right], \boldsymbol{u} \in M_C\end{aligned}$$

With our operators defined we can now construct the iterative relation that will be used to maximization our objective functional (3.32)

$$\boldsymbol{\eta}_{\alpha}^{(n+1)} = \mathbf{R}\left[\boldsymbol{\eta}_{\alpha}^{(n)} + \tau_n \boldsymbol{d}^{(n)}\right]. \tag{3.52}$$

To simply notation, let $\boldsymbol{G}^{(n)} := \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}^{(n)})$ and let $\mathcal{T}_n := \mathcal{T}_{\boldsymbol{\eta}_{\alpha}^{(n)}}\mathcal{M}_C$. The search direction at iteration n ($\boldsymbol{d}^{(n)}$) is computed as

$$\begin{aligned}\boldsymbol{d}^{(0)} &= P_{\mathcal{T}_0}\boldsymbol{G}^{(0)}\\ \boldsymbol{d}^{(n)} &= P_{\mathcal{T}_n}\boldsymbol{G}^{(n)} + \beta_n \Gamma_{\tau_{n-1}\boldsymbol{d}^{(n-1)}}(\boldsymbol{d}^{(n-1)}), \quad n \geq 1,\end{aligned}$$

where the 'momentum' term is computed using the Polak-Ribière approach [24].

$$\beta_n = \frac{\left\langle P_{\mathcal{T}_n}\boldsymbol{G}^{(n)}, \left(P_{\mathcal{T}_n}\boldsymbol{G}^{(n)} - \Gamma_{\tau_{n-1}\boldsymbol{d}^{n-1}}P_{\mathcal{T}_{n-1}}\boldsymbol{G}^{(n-1)}\right)\right\rangle_{G^{\sigma}}}{\|P_{\mathcal{T}_{n-1}}\boldsymbol{G}^{(n-1)}\|^2_{G^{\sigma}}}. \tag{3.54}$$

Finally our optimal step size is determined by the distance along $\boldsymbol{d}^{(n)}$ which maximizes the objective functional, which is computed using Brent's method.

$$\tau_n = \underset{\tau>0}{\operatorname{argmax}}\, \Phi_T\left(\mathbf{R}\left[\eta^{(n)} + \tau\boldsymbol{d}^{(n)}\right]\right) \tag{3.55}$$

4 Numerical Methods

4.1 Spatial Discretization & Spectral Methods

Simulations were computed using a standard pseudospectral fourier method on a periodic unit cube, discretized in a uniform grid of spatial resolutions $N = 128^3, 256^3, 512^3, 1024^3$. Derivatives are evaluated in the Fourier space and nonlinear products are computed in the physical space. The standard 2/3 dealiasing rule is also applied. Our interest is in the limit $\alpha \to 0$. Therefore, we use the regularization parameters $\alpha = 2^{-6}, 2^{-7}, 2^{-8}, 2^{-9}, 2^{-10}$.

4.2 Time Stepping

Time stepping is performed with the explicit fourth-order Runge-Kutta (RK4) method. The determination of our time step size Δt is based on three factors. First, we consider the numerical stability of the models. Second, we consider the accuracy of the solution, determined by the relative error of the objective functional (3.32) to the results for a minimum step size. Finally, we consider the conservation of the α-energy (1.6), determined by the maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$. These factors are analysed for each resolution and value of α for the Taylor Green Vortex.

The goal of our optimization problem is to identify initial conditions $\tilde{\boldsymbol{\eta}}_{\alpha,T}$ that produce 'extreme' behaviour in solutions to the Euler-Voigt equations. Therefore, the Taylor Green Vortex will by definition be the least extreme initial condition examined. To ensure stability, accuracy, and conservation of α-energy are maintained upon optimization we must select a step size smaller than the minimum step size identified by these criteria.

4.2.1 Numerical Stability & Accuracy

We begin with the stability and accuracy of the Euler-Voigt model. To determine the stability and accuracy of the model at each step size we start from the iteration zero initial conditions and evaluate the objective functional (3.32) for the time window $T = 5$. First we do this for $\Delta t = 2^{-8}$ and designate the result as our 'correct' value Φ_c. We then compute the relative error of the objective functional for each value of Δt (Φ_{dt}) to Φ_c

$$\epsilon_{rel} = \left| \frac{\Phi_c - \Phi_{dt}}{\Phi_c} \right|. \tag{4.56}$$

For models that experienced numerical instability in the time window $[0, T]$ we set ϵ_{rel} to infinity.

Tables (4.1) to (4.4) show the results for ϵ_{rel} at the four resolutions utilized in this analysis. The tables shows that the accuracy of the objective functional decreases as $\alpha \to 0$ and increases as $\Delta t \to 0$. In addition, the accuracy of the model overall increases as the resolution increases. Finally, the size of the instability region does increase with resolution, but only for the smallest value of α. For all other values of α, the stability is unaffected over the time window $t \in [0, T]$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.0045	7.00E-05
2^{-3}	Inf	Inf	0.0572	0.0011	3.64E-06
2^{-4}	0.1662	0.1157	0.0332	4.85E-04	2.02E-07
2^{-5}	0.1093	0.0715	0.0181	2.33E-04	1.21E-08
2^{-6}	0.0593	0.0369	0.0085	1.02E-04	8.63E-10
2^{-7}	0.0228	0.0137	0.003	3.45E-05	9.44E-11
2^{-8}	0	0	0	0	0

Table 4.1: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 128^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.0112	1.82E-04	3.64E-06
2^{-4}	Inf	0.0652	0.0021	9.46E-06	2.01E-07
2^{-5}	0.1097	0.0348	7.13E-04	5.12E-07	1.17E-08
2^{-6}	0.0619	0.0171	3.08E-04	3.00E-08	7.01E-10
2^{-7}	0.025	0.0157	1.05E-04	1.96E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 4.2: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 256^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.009	1.82E-04	3.64E-06
2^{-4}	Inf	0.0174	7.61E-04	9.45E-06	2.01E-07
2^{-5}	Inf	0.0036	3.63E-05	5.10E-07	1.17E-08
2^{-6}	0.0306	7.85E-04	1.72E-06	2.90E-08	7.01E-10
2^{-7}	0.0115	2.42E-04	8.42E-08	1.63E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 4.3: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 512^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.009	1.82E-04	3.64E-06
2^{-4}	Inf	0.015	7.61E-04	9.45E-06	2.01E-07
2^{-5}	Inf	0.0022	3.64E-05	5.10E-07	1.17E-08
2^{-6}	0.0039	2.64E-04	1.71E-06	2.90E-08	7.01E-10
2^{-7}	5.17E-04	4.92E-06	8.36E-08	1.63E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 4.4: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 1024^3$ and time window $T = 5$.

4.2.2 Conservation of α-Energy

The conservation of α-energy is also used to verify the model for a given resolution, regularization parameter, and time window. The RK4 method does not conserve energy, and so the α-energy of our model gradually decreases over time. The tables below show the maximum relative error of the α-energy for each α and Δt,

$$\epsilon_\alpha = \max_{t\in[0,T]} \left| \frac{E_\alpha(t) - E_\alpha(0)}{E_\alpha(t)} \right|. \tag{4.57}$$

For models that experienced numerical instability in the time window $[0, T]$ we set ϵ_α to infinity.

The Tables show that the conservation of α-energy error ϵ_α increases as $\alpha \to 0$ and decreases as $\Delta t \to 0$. In addition, the error decreases as the resolution increases.

Based on the results of the stability, accuracy, and α-energy conservation we can now select the appropriate time step size for each α and resolution, shown in Table (4.9). These step sizes are selected because they are the largest step sizes that ensure

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	3.82E-04	8.73E-06
2^{-3}	Inf	Inf	1.94E-03	6.91E-05	3.72E-07
2^{-4}	3.38E-03	2.74E-03	1.11E-03	3.05E-05	1.76E-08
2^{-5}	2.38E-03	1.81E-03	6.49E-04	1.57E-05	9.59E-10
2^{-6}	1.51E-03	1.09E-03	3.56E-04	8.05E-06	7.55E-11
2^{-7}	8.67E-04	6.06E-04	1.87E-04	4.08E-06	1.47E-11
2^{-8}	4.69E-04	3.21E-04	9.59E-05	2.05E-06	6.00E-12

Table 4.5: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 128^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	4.34E-04	1.49E-05	3.72E-07
2^{-4}	Inf	8.36E-04	5.61E-05	6.85E-07	1.75E-08
2^{-5}	9.63E-04	4.35E-04	1.80E-05	3.40E-08	9.09E-10
2^{-6}	6.33E-04	2.48E-04	8.97E-06	1.89E-09	4.95E-11
2^{-7}	3.83E-04	1.37E-04	4.59E-06	1.38E-10	2.89E-12
2^{-8}	2.14E-04	7.23E-05	2.33E-06	2.37E-11	2.95E-12

Table 4.6: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 256^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.72E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.89E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.58E-04	8.18E-06	5.00E-08	1.82E-09	4.95E-11
2^{-7}	9.02E-05	3.74E-06	2.47E-09	1.03E-10	3.44E-12
2^{-8}	4.98E-05	1.91E-06	1.42E-10	6.31E-12	2.69E-12

Table 4.7: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 512^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.53E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.67E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.46E-05	1.26E-06	4.99E-08	1.82E-09	4.95E-11
2^{-7}	2.53E-06	5.12E-08	2.45E-09	1.03E-10	3.25E-12
2^{-8}	1.06E-06	2.35E-09	1.30E-10	6.31E-12	2.78E-12

Table 4.8: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 1024^3$ and time window $T = 5$.

N \ α	1/1024	2/1024	4/1024	8/1024	16/1024
128^3	2^{-8}	2^{-8}	2^{-6}	2^{-5}	2^{-3}
256^3	2^{-8}	2^{-8}	2^{-6}	2^{-4}	2^{-3}
512^3	2^{-8}	2^{-7}	2^{-5}	2^{-4}	2^{-3}
1024^3	2^{-8}	2^{-7}	2^{-5}	2^{-4}	2^{-3}

Table 4.9: Selection of Δt for each resolution and α.

the stability, accuracy and α-energy conservation of the model from the Taylor Green Vortex.

4.3 Determination of Gevrey class size

In Section (2.2) we introduce the Gevrey space in which we search for initial conditions (2.21). The size of the space is determined by two variables, $s \geq 3$ and $\sigma > 0$. We use a fixed value of $s = 3$. In order to determine the value of σ we start with the optimization problem for $\alpha = 16/1024$, $T = 5$, and resolution $N = 128^3$. The problem is repeated for $\sigma =$ 1E-1, 1E-2, 1E-3, 1E-4, 1E-5.

4.4 Spectrum Validation

The energy spectrum of the solution $\boldsymbol{u}_\alpha(\boldsymbol{x}, t)$ is defined as,

$$e(k,t) := \frac{1}{2} \sum_{j \leq |\boldsymbol{j}| < j+1} |\hat{\boldsymbol{u}}_{\alpha \boldsymbol{j}}(t)|^2, \quad k = 2\pi j, \quad j \in \mathbb{N}. \tag{4.58}$$

The regularization term of (1.4) acts as a filter which suppresses the growth of small-scale structures of length-scale smaller than the regularization parameter α. Therefore, the spectrum (4.58) is suppressed for wavenumbers $k > k_\alpha$ where

$$k_\alpha := \frac{2\pi}{\alpha}. \tag{4.59}$$

Due to this filter, the solution to (1.4) can remain well resolved. This is ideal for our analysis because it ensures that the model fully captures the small scale structures of the fluid. When this occurs we can be certain that our objective functional (3.32) is accurate. We begin by examining the spectrum of the solution from the Taylor Green Vortex on the time window $[0, T]$ for $T = 5$.

The plots () to () show the semi-log energy vs wavenumber graph of the solution over time.

initial condition shown below demonstrates the exponential decay expected by a analytic function. As the solution evolves in time the rate of decay decreases due to aliasing. While this can be reduced by anti-aliasing techniques, it cannot be entirely prevented

Low resolution and small alpha simulations are especially vulnerable to aliasing errors and the solutions stop being analytic before the chosen timescale.

The chosen timescales, alphas, and resolutions are validated in the graphs below which show that the spectral decay remains exponential and thus the solutions remain analytic.

4.5 Kappa Test Validation

The simulations are discretized in space using a psuedospectral method and time using a fourth order Runge Kutta method. The kappa test is a means of validating these discretization methods based on the validation method of [**?**]. We define the value κ as

$$\kappa(\epsilon) = \frac{\epsilon^{-1}\left[\Phi_{\alpha,T}(\boldsymbol{\eta} + \epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})\right]}{\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}'\rangle} \tag{4.60}$$

where $\Phi_{\alpha,T}$ is the objective functional (3.32), $\boldsymbol{\eta}$ is the initial condition, and $\nabla\Phi_{\alpha,T}$ is the gradient of the objective functional (3.45). The numerator is the first order finite difference approximation of the Gateaux differential. The denominator is the Reisz form of the differential that utilizes the adjoint system (**??**). If the numerical approximation methods are accurate, then we expect $\kappa \approx 1$.

A log-log plot of $|\kappa - 1|$ vs ϵ can be broken up into three regions, each dominated by different error sources. These are the discretization errors, round off errors, and truncation errors. The round off errors are caused by finite precision numerical representation and so are approximately of order $O(\epsilon^{-1})$. Therefore, we expect the value of κ to be dominated by round off errors for small ϵ values. The truncation errors are caused by using a first order finite difference formula to calculate the derivative. The truncation errors are of order $O(\epsilon)$. Therefore, we expect the value of κ to be dominated by the truncation errors for large values of ϵ. Between these two regions the value of κ will be determined by the discretization errors in the numerical solutions, which are themselves a product of the spatial and temporal resolution. With smaller values of Δx and Δt causing the error to decrease. This will cause the value of $\log_{10} |kappa - 1|$ to decrease as the resolution increases. Additionally, an increase in the resolution will decrease the range of ϵ over which the discretization errors dominate. Therefore, we expect the plot of κ vs ϵ to be of the general form shown in the graph below.

The following κ test graphs were generated using a fixed spacial resolution of 128 and with varying temporal resolutions $\Delta t \in [2^{-4}, 2^{-12}]$. The initial conditions used are the 3D Taylor Green, 3D ABC, and 3D ABC Mixture, each with a random initial perturbation. The simulations were run for timescales of $T = 1$ and $T = 3$. The results are shown below.

The results from the 3D Taylor-Green κ test show the expected behaviour from the theoretical kappa test graph. The value of kappa increases at very small and large values of epsilon. The value of kappa approaches 1 as Δt decreases, and so the plateau region lowers and decreases in width. The T=1 time scale results follow very closely to the expected results while the T=3 time scale simulations show deviations from the the expected behaviour. These differences are due to the random initial direction of each simulation having more time to deviate from one another.

The kappa test results for the 3D ABC initial condition exhibit the same lowering of kappa as the time step decreases. However, unlike the Taylor Green initial condition, the results from the 3D ABC κ test have little to no plateau region. This indicates that the 3D ABC initial condition simulations are much more sensitive to the round off and truncation errors.

The results from the 3D ABC Mixture κ test show the same expected behaviour as the 3D Taylor Green kappa test.

The three kappa test results align well with the expected theoretical result. Therefore, we conclude that the discretization and time stepping techniques used in this analysis are valid and accurate.

5 Results

Theorem 2.1.3 provides the criteria we use to confirm that singularity formation occurs in Euler as $\alpha \to 0$. First we check that condition (2.8) is satisfied, then we check that (**??**) is satisfied. It is not necessarily clear whether or not the value used in each criterion is converging to zero or to some small positive value. In order to make this determination we use the method of [17], which proposes the following ansatz,

$$\sup_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2} \approx \mathcal{O}(\alpha^p), \tag{5.61}$$

for $T^* > 0$ sufficiently large and for some power p. From Theorem 2.1.3, if there exists a finite time $T^* > 0$ such that

$$\limsup_{\alpha\to 0} \left(\alpha^2 \sup_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0. \tag{5.62}$$

then $\boldsymbol{u}$ develops a singularity in the interval $[0, T^*]$. From this ansatz, in order for singularity formation to occur, we must have $p \leq -1$.

We present numerical evidence that singularity formation does not occur from the Taylor Green Vortex as $\alpha \to 0$. In addition, we present numerical evidence that upon optimization ... RESULTS!

5.0.1 Pre-Optimization Results

We begin by presenting our results for the TGV without optimization, i.e., $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$.

First show that condition (2.8) is satisfied.

Results of $\alpha^2\Phi_{\alpha,T}$ for TGV

Without optimization $\boldsymbol{\eta}$ and $\tilde{\boldsymbol{\eta}}_{\alpha,T}$, for all values of α, are simply the TGV. Therefore, it is trivially true that condition (2.8) is satisfied. So we only need to examine the value of p as $\alpha \to 0$ to determine if a singularity forms in Euler.

Comparison to Larios et al. 2018

Compare our results to Larios et al. 2018. Our results conflict with those of Larios et al. 2018. Their analysis found that blow-up does occur for the Taylor Green Vortex with a $p \leq -1$. Additionally the minimum slope occurs at $T^* \approx 4$ which coincides with those of Bustamante and Brachet. However, we believe their analysis improperly scales the Taylor Green Vortex. RESULTS RESULTS RESULTS.

Results of $\alpha^2\Phi_{\alpha,T}$ for INIT1

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT1})$ as $\alpha \to 0$

Results of α slope for INIT1

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for INIT2

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT2})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT2

Results of α slope as $\alpha \to 0$

5.0.2 Optimization Results

We present the Optimization Results for each initial condition

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for TGV

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{TGV})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for TGV

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for INIT1

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT1})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT1

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for INIT2

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT2})$ as $\alpha \to 0$

Results of α slope for INIT2

Results of α slope as $\alpha \to 0$

6 Conclusion

Every thesis also needs a concluding chapter

A Function Spaces

In this analysis we use the Lebesgue space $L^p(\Omega)$ which consists of real-valued measurable functions $\boldsymbol{u}$, defined on Ω, and $p \in \mathbb{R}^+$ for which

$$\int_\Omega |\boldsymbol{u}|^p \, d\boldsymbol{x} < \infty.$$

For $L^p(\Omega)$ the norm is defined as

$$\|\boldsymbol{u}\|_{L^p} = \left(\int_\Omega |\boldsymbol{u}|^p \, d\boldsymbol{x} \right)^{1/p},$$

where $1 \leq p < \infty$. For $p = \infty$, the norm is given by

$$\|\boldsymbol{u}\|_{L^\infty} = \operatorname*{ess\,sup}_{\boldsymbol{x} \in \Omega} |\boldsymbol{u}|,$$

where ess sup is the essential supremum of $|\boldsymbol{u}|$ on Ω [12].

For any nonnegative integer k, the space $C^k(\Omega)$ is the space of functions $\boldsymbol{u}$ which, together with all partial derivatives $\partial^\alpha \boldsymbol{u}$ of orders $|\alpha| \leq k$, are continuous on Ω [12].

For $s \in \mathbb{R}^+$ the L^2-based Sobolev space $H^s(\Omega)$ is the space of all functions for which the finite norm $\|\cdot\|_{H^s} < \infty$ defined by

$$\|\boldsymbol{u}\|_{H^s}^2 := (2\pi)^3 \sum_{k\in\mathbb{Z}^3} (1+|k|^{2s})|\hat{\boldsymbol{u}}_k|^2,$$

where $\hat{\boldsymbol{u}}_k$ is the Fourier coefficient [26]. For any function $\boldsymbol{u} \in H^s(\mathbb{T}^3)$ the $\dot{H}^s$ seminorm is defined by

$$\|\boldsymbol{u}\|_{\dot{H}^s}^2 := (2\pi)^3 \sum_{k\in\dot{\mathbb{Z}}^3} |k|^{2s}|\hat{\boldsymbol{u}}_k|^2$$

B Derivation of the Gateaux Directional Differential

The equation for the Gateaux differential (3.36) is

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\Phi_{\alpha,T}(\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})].$$

The objective functional (3.32) is defined as,

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{L^2} = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}.$$

Substituting the second definition of (3.32) into (3.36),

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}')\|^2_{\dot{H}^1} - \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}].$$

The equation for the $\dot{H}^1$ seminorm is

$$\|u\|^2_{\dot{H}^1} = (2\pi)^3 \sum_{k\in\mathbb{Z}^3} |k|^2 |\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2.$$

Substituting in the equation for the $\dot{H}^1$ seminorm

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}} + \epsilon\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|^2 - (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2].$$

This simplifies to

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2 2\epsilon|\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}| + (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2\epsilon^2|\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|^2].$$

Taking the limit $\epsilon \to 0$

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|.$$

This can be expressed in the integral form

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\int_{\mathbb{T}^3}\sum_{k\in\mathbb{Z}^3}|k|^2\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}e^{2ikx}dx.$$

Which is equivalent to the inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}_{\boldsymbol{k}}e^{ikx}, \sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}'_{\boldsymbol{k}}e^{ikx}\right\rangle_{L^2}.$$

This gives us the Gateaux differential in the form of the $\dot{H}^1$ inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\boldsymbol{u}_\alpha(\cdot,T), \boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1}.$$

The integral form of the $\dot{H}^1$ inner product is

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle_{\dot{H}^1} = \int_{\mathbb{T}^3} |D|\boldsymbol{u} \cdot |D|\boldsymbol{v} \, dx,$$

where $|D|$ is the operator defined in (2.24). Therefore, the Gateaux differential can be expressed as

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = 2 \int_{\mathbb{T}^3} |D|\boldsymbol{u}_\alpha(x,T) \cdot |D|\boldsymbol{u}'_\alpha(x,T) \, dx,$$

which can be rearranged to the form

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = \int_{\mathbb{T}^3} 2|D|^2 \boldsymbol{u}_\alpha(x,T) \cdot \boldsymbol{u}'_\alpha(x,T) \, dx.$$

We now make use of the adjoint system terminal condition derived in Appendix (D)

$$\boldsymbol{u}^*_\alpha(x,T) = 2|D|^2 \boldsymbol{u}_\alpha(x,T).$$

Therefore

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{u}^*_\alpha(x,T) \cdot \boldsymbol{u}'_\alpha(x,T) \, dx.$$

Now using the duality-pairing defined in Appendix (D)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{\eta}'(x) \cdot \boldsymbol{u}^*_\alpha(x,0) \, dx.$$

This is the inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = \langle \boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'(x) \rangle_{L^2}.$$

Therefore,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T), \boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} = \left\langle \boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'(x)\right\rangle_{L^2}.$$

C Linearization of the Euler-Voigt System

Euler-Voigt Equations,

$$\begin{cases} -\alpha^2 \partial_t \Delta \boldsymbol{u}_\alpha + \partial_t \boldsymbol{u}_\alpha + (\boldsymbol{u}_\alpha \cdot \boldsymbol{\nabla}) \boldsymbol{u}_\alpha + \boldsymbol{\nabla} p = 0 \\ \boldsymbol{\nabla} \cdot \boldsymbol{u}_\alpha = 0 \\ \boldsymbol{u}_\alpha \big|_{t=0} = \boldsymbol{\eta}. \end{cases}$$

Substituting in $\boldsymbol{u}_\alpha = \boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha$, $\boldsymbol{\eta} = \boldsymbol{\eta} + \boldsymbol{\eta}'$, and $p = p + p'$,

$$\begin{cases} -\alpha^2 \partial_t \Delta (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + \partial_t (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + ((\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) \cdot \boldsymbol{\nabla})(\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + \boldsymbol{\nabla}(p + p') = 0 \\ \boldsymbol{\nabla} \cdot (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) = 0 \\ (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha)\big|_{t=0} = \boldsymbol{\eta} + \boldsymbol{\eta}'. \end{cases}$$

Expanding

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}_\alpha - \alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \\ \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{\nabla}p + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot(\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) = 0 \\ (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha)|_{t=0} = \boldsymbol{\eta} + \boldsymbol{\eta}'. \end{cases}$$

Substituting in the Euler-Voigt equations we simplify to

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases}$$

Linearizing the system by removing non-linear terms, we derive the Linearized Euler-Voigt system

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases}$$

This can be rearranged to the form

$$\begin{cases} \partial_t\boldsymbol{u}'_\alpha + (I - \alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p') = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases} \tag{C.63}$$

D Derivation of the Adjoint System

For this derivation we will exclude the α subscript on the u_α, u'_α, and u^*_α terms for the sake of simplicity. Additionally, we substitute the integration with the following notation

$$\int_0^T \int_{\mathbb{T}^3} u\, dt = \int_D u\, dt$$

We derive the adjoint system using the following duality pairing relation,

$$(L[u'], u^*) := \int_D L[u']u^* dxdt = (u', L[u^*]) + \int_a^b u'u^*|_{t=T} - \int_a^b u'u^*|_{t=0} dx = 0.$$

Substituting in the equation from the Euler-Voigt perturbation system,

$$(L[u'], u^*) := \int_D (\partial_t u' + (I - \alpha^2 \Delta)^{-1}(u.\nabla u' + u' \cdot \nabla u_+ \nabla p'))u^* + (\nabla \cdot u')p^* dx\, dt.$$

This integral can be split into five parts. Solving the first part with integration by parts, with $u = u^*$ and $v' = \partial_t u'$

$$\int_D u^* \partial_t u' dx\, dt = \int_{\mathbb{T}^3} u^* u'|_{t=0}^{t=T} dx - \int_D u' \partial_t u^* dx\, dt.$$

Starting with the second part and expressing it in Einstein Summation notation,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}u_j\frac{\partial u_i'}{\partial x_j}\right]u_i^* dx\,dt.$$

Using the derivative product rule and the fact that $\nabla\cdot u = 0$,

$$\frac{\partial}{\partial x_j}(u_j u_i') = \frac{\partial u_j}{\partial x_j}u_i' + u_j\frac{\partial u_i'}{\partial x_j} = u_j\frac{\partial u_i'}{\partial x_j}.$$

Substituting this into the integral,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}\frac{\partial}{\partial x_j}(u_j u_i')\right]u_i^* dx\,dt.$$

Using the periodic boundaries of the system, we can now apply integration by parts with $u = [(I-\alpha^2\Delta)^{-1}u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u_j u_i')$,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = -u_j u_i'\frac{\partial}{\partial x_j}\left[(I-\alpha^2\Delta)^{-1}u_i^*\right]dx\,dt.$$

Converting back to vector notation,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = -u'\cdot\left[u\cdot\nabla\left[(I-\alpha^2\Delta)^{-1}u^*\right]\right]dx\,dt.$$

Starting with the third part and expressing it in Einstein Summation notation,

$$\left[(I-\alpha^2\Delta)^{-1}u'\cdot\nabla u\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}u_j'\frac{\partial u_i}{\partial x_j}\right]u_i^* dx\,dt.$$

Applying integration by parts with $u = [(I-\alpha^2\Delta)^{-1}u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u_j' u_i)$

$$\left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u_j' u_i \frac{\partial}{\partial x_j}\left[(I-\alpha^2\Delta)^{-1}u_i^*\right] dx\, dt.$$

Because the goal is to extract u' and $v \cdot w = v^T \cdot w^T$, we can apply a transpose to the above equation and so swap the indices to,

$$\left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u_i' u_j \frac{\partial}{\partial x_i}\left[(I-\alpha^2\Delta)^{-1}u_j^*\right] dx\, dt.$$

Converting back to vector notation,

$$\left[(I-\alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u' \cdot \left[u \cdot \left[\nabla(I-\alpha^2\Delta)^{-1}u^*\right]^T\right] dx\, dt.$$

Starting with the fourth part of the integral and expressing it in Einstein Summation notation,

$$\nabla p' \cdot u^* dx\, dt = \frac{\partial p'}{\partial x_i} u_i^* dx\, dt.$$

Applying integration by parts with $u = u_i^*$ and $v' = \partial x_i p'$,

$$\nabla p' \cdot u^* dx\, dt = -p' \frac{\partial u_i^*}{\partial x_i} dx\, dt.$$

Converting back to vector notation,

$$\nabla p' \cdot u^* dx\, dt = -p'(\nabla \cdot u^*) dx\, dt.$$

Starting with the final part of the integral and expressing it in Einstein Summation notation,

$$(\nabla \cdot u')p^* dx\, dt = \frac{\partial u_i'}{\partial x_i} p^* dx\, dt.$$

Applying integration by parts with $u = p^*$ and $v' = \partial x_i u_i'$,

$$(\nabla \cdot u')p^* dx\, dt = -\frac{\partial p^*}{\partial x_i} u_i' dx\, dt.$$

Converting back to vector notation,

$$(\nabla \cdot u')p^* dx\, dt = -u' \cdot \nabla p^* dx\, dt.$$

Combining the integrals,

$$\int_{\mathbb{T}^3} u^* u'|_{t=0}^{t=T} dx - u' \left[\partial_t u^* - \left[u \cdot \nabla \left[(I - \alpha^2 \Delta)^{-1} u^*\right]\right] - \left[u \cdot \left[\nabla (I - \alpha^2 \Delta)^{-1} u^*\right]^T\right] - \nabla p^*\right] - p'(\nabla \cdot u^*)$$

This gives us our adjoint system,

$$\mathbb{L}^* \begin{bmatrix} u_\alpha^* \\ \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t u^* - (\nabla (I - \alpha^2)^{-1} u_\alpha^* + (\nabla (I - \alpha^2)^{-1} u_\alpha^*)^T) u_\alpha - \nabla p_\alpha^* \\ \\ -\nabla \cdot u_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ \\ 0 \end{bmatrix}$$

To derive the terminal condition of the adjoint system we compare the second term of the duality pairing to the objective functional.

$$\int_{\mathbb{T}^3} u'(x,T) \boldsymbol{u}^*(x,T) dx = 2 \left\langle u(x,T), u'(x,T) \right\rangle_{\dot{H}^1}$$

$$= 2\int_{\mathbb{T}^3} |D|u(x,T)\boldsymbol{u}'(x,T)dx$$
$$= \int_{\mathbb{T}^3} 2|D|^2 u(x,T)\boldsymbol{u}'(x,T)dx$$

Therefore,

$$\boldsymbol{u}^*(x,T) = 2|D|^2 u(x,T)$$

E Proof of Relation (2.21)

Proof. Let σ_1, σ_2 be defined such that

$$0 < \sigma_1 < \sigma_2.$$

Therefore, from (2.22)

$$\begin{aligned}
\|\boldsymbol{v}\|_{G^{\sigma_2}} &= \left[\int_{\mathbb{T}^3} (1+|D|^2)^s e^{2\sigma_2|D|}\boldsymbol{v}\cdot\boldsymbol{v}\,d\boldsymbol{x}\right]^{1/2} \\
&> \left[\int_{\mathbb{T}^3} (1+|D|^2)^s e^{2\sigma_1|D|}\boldsymbol{v}\cdot\boldsymbol{v}\,d\boldsymbol{x}\right]^{1/2} = \|v\|_{G^{\sigma_1}} \\
&> \left[\int_{\mathbb{T}^3} (1+|D|^2)^s e^{0}\boldsymbol{v}\cdot\boldsymbol{v}\,d\boldsymbol{x}\right]^{1/2} = \|\boldsymbol{v}\|_{G^0} \\
&= \left[\int_{\mathbb{T}^3} (1+|D|^2)^s \boldsymbol{v}\cdot\boldsymbol{v}\,d\boldsymbol{x}\right]^{1/2} = \|v\|_{H^s}
\end{aligned} \tag{E.64}$$

Thus,

$$\|\boldsymbol{v}\|_{G^{\sigma_2}} > \|\boldsymbol{v}\|_{G^{\sigma_1}} > \|\boldsymbol{v}\|_{G^0}.$$

It follows from this that if $\|\boldsymbol{v}\|_{G^{\sigma_2}} < \infty$ then $\|\boldsymbol{v}\|_{G^{\sigma_1}} < \infty$ and so if $v \in G^{\sigma_2}$ then $v \in G^{\sigma_1}$. However, the converse is not true. The same argument shows that if $v \in G^{\sigma_1}$

then $v \in G^0$ which is equivalent to $v \in H^s$. Therefore,

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s.$$

$\square$

F Evaluation of $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ for the General Taylor Green Vortex

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is given in (2.29) To simplify the notation we drop the subscript TGV and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The formula for $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ is

$$\|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla\boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^{3}\int_\Omega (\partial_i\boldsymbol{\eta}_j)^2\, d\boldsymbol{x}\right]^{1/2},$$

where ∂_i is the partial derivative with respect to the i-th component and $\boldsymbol{\eta}_j$ is the j-th component of $\boldsymbol{\eta}$. Because the 3rd component of $\boldsymbol{\eta}$ is zero, the partial derivative is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned}\sum_{i,j=1}^{3}\int_\Omega (\partial_i\boldsymbol{\eta}_j)^2\, d\boldsymbol{x} &= \int_\Omega (\partial_1\boldsymbol{\eta}_1)^2\, d\boldsymbol{x} + \int_\Omega (\partial_2\boldsymbol{\eta}_1)^2\, d\boldsymbol{x} + \int_\Omega (\partial_3\boldsymbol{\eta}_1)^2\, d\boldsymbol{x} \\ &\quad + \int_\Omega (\partial_1\boldsymbol{\eta}_2)^2\, d\boldsymbol{x} + \int_\Omega (\partial_2\boldsymbol{\eta}_2)^2\, d\boldsymbol{x} + \int_\Omega (\partial_3\boldsymbol{\eta}_2)^2\, d\boldsymbol{x}.\end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned}
\sum_{i,j=1}^{3}\int_\Omega (\partial_i \boldsymbol{\eta}_j)^2\,d\boldsymbol{x} = &\int_\Omega (2\pi V_0/L\cos(x')\cos(y')\cos(z'))^2\,d\boldsymbol{x}\\
&+\int_\Omega (-2\pi V_0/L\sin(x')\sin(y')\cos(z'))^2\,d\boldsymbol{x}\\
&+\int_\Omega (-2\pi V_0/L\sin(x')\cos(y')\sin(z'))^2\,d\boldsymbol{x}\\
&+\int_\Omega (2\pi V_0/L\sin(x')\sin(y')\cos(z'))^2\,d\boldsymbol{x}\\
&+\int_\Omega (-2\pi V_0/L\cos(x')\cos(y')\cos(z'))^2\,d\boldsymbol{x}\\
&+\int_\Omega (2\pi V_0/L\cos(x')\sin(y')\sin(z'))^2\,d\boldsymbol{x},
\end{aligned}$$

which simplifies to

$$\begin{aligned}
\sum_{i,j=1}^{3}\int_\Omega (\partial_i \boldsymbol{\eta}_j)^2\,d\boldsymbol{x} = &\,4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\cos^2(y')\cos^2(z')\,d\boldsymbol{x}\\
&+4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\sin^2(y')\cos^2(z')\,d\boldsymbol{x}\\
&+4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\cos^2(y')\sin^2(z')\,d\boldsymbol{x}\\
&+4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\sin^2(y')\cos^2(z')\,d\boldsymbol{x}\\
&+4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\cos^2(y')\cos^2(z')\,d\boldsymbol{x}\\
&+4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\sin^2(y')\sin^2(z')\,d\boldsymbol{x}.
\end{aligned}$$

Because the integrand is separable, the integrals can be reduced to products of,

$$\int_0^L \cos^2(x')\,dx = \frac{L}{2}, \qquad \text{and} \qquad \int_0^L \sin^2(x')\,dx = \frac{L}{2}.$$

Therefore, each of the six integrals is equal to

$$\frac{4\pi^2 V_0^2}{L^2}\frac{L^3}{8}.$$

The sum of the six integrals is

$$\sum_{i,j=1}^{3}\int_\Omega (\partial_i \boldsymbol{\eta}_j)^2\, d\boldsymbol{x} = 6\frac{4\pi^2 V_0^2}{L^2}\frac{L^3}{8} = 3\pi^2 V_0^2 L.$$

Substituting into the formula for $\|\cdot\|_{\dot{H}^1}$ gives

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \left[3\pi^2 V_0^2 L\right]^{1/2} = \sqrt{3L}\pi V_0.$$

G Evaluation of $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ for the General Taylor Green Vortex

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is given in (2.29). To simplify the notation we drop the subscript TGV and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The vorticity of $\boldsymbol{\eta}$ is

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} \partial_2 \boldsymbol{\eta}_3 - \partial_3 \boldsymbol{\eta}_2 \\ \\ \partial_3 \boldsymbol{\eta}_1 - \partial_1 \boldsymbol{\eta}_3 \\ \\ \partial_1 \boldsymbol{\eta}_2 - \partial_2 \boldsymbol{\eta}_1. \end{bmatrix}$$

where ∂_i is the partial derivative with respect to the i-th component and $\boldsymbol{\eta}_j$ is the j-th component of $\boldsymbol{\eta}$. Substituting in the partial derivatives

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} 0 - (-(2\pi V_0/L)\cos(x')\sin(y')\sin(z') \\ \\ -(2\pi V_0/L)\sin(x')\cos(y')\sin(z') - 0 \\ \\ (2\pi V_0/L)\sin(x')\sin(y')\cos(z') + (2\pi V_0)\sin(x')\sin(y')\cos(z'), \end{bmatrix}$$

which simplifies to

$$\nabla \times \eta_{TG} = \begin{bmatrix} (2\pi V_0/L)\cos(x')\sin(y')\sin(z') \\ \\ -(2\pi V_0/L)\sin(x')\cos(y')\sin(z') \\ \\ (4\pi V_0/L)\sin(x')\sin(y')\cos(z') \end{bmatrix}.$$

The $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \boldsymbol{\eta}$.

$$\begin{aligned} \|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = \max(&|(2\pi V_0/L)\cos(x')\sin(y')\sin(z')| \\ &, |-(2\pi V_0/L)\sin(x')\cos(y')\sin(z')| \\ &, |(4\pi V_0/L)\sin(x')\sin(y')\cos(z')|) \end{aligned}$$

The maximums are

$$\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = \max((2\pi V_0/L), (2\pi V_0/L), (4\pi V_0/L))$$

Therefore,

$$\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 4\pi V_0/L.$$

H Rescaling the Euler equations

Given a solution $\boldsymbol{u}$ of the Euler equations (1.2), we construct a new scaled solution with

$$\tau = \lambda t, \qquad \text{and} \qquad \boldsymbol{y} = \beta \boldsymbol{x}.$$

where $\lambda, \beta > 0$ are constants. Our new solution $\boldsymbol{v}$ is defined as

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{y}, \tau) = \gamma \boldsymbol{u}(\boldsymbol{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t \boldsymbol{u} = \frac{1}{\gamma} \frac{\partial \boldsymbol{v}}{d\tau} \frac{d\tau}{dt} = \frac{\lambda}{\gamma} \partial_\tau \boldsymbol{v}.$$

and

$$\boldsymbol{\nabla} \boldsymbol{u} = \frac{1}{\gamma} \boldsymbol{\nabla} \boldsymbol{v} \frac{d\boldsymbol{y}}{d\boldsymbol{x}} = \frac{\beta}{\gamma} \boldsymbol{\nabla} \boldsymbol{v}.$$

Substituting $\boldsymbol{v}$ into equation (1.2a)

$$\frac{\lambda}{\gamma} \partial_\tau \boldsymbol{v} + \frac{\beta}{\gamma^2} (\boldsymbol{v} \cdot \boldsymbol{\nabla}) \boldsymbol{v} + \boldsymbol{\nabla} q = 0,$$

where q is defined as whatever pressure is required to satisfy the incompressibility condition. This can be rearranged to the form,

$$\frac{\lambda\gamma}{\beta}\partial_\tau \boldsymbol{v} + (\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v} + \boldsymbol{\nabla} q' = 0.$$

Therefore, our function $\boldsymbol{v}$ can only be a solution to the Euler equations if

$$\frac{\lambda\gamma}{\beta} = 1.$$

For the Taylor Green Vortex with $L = 2\pi$ and $V_0 = 1$, let

$$\gamma = \lambda = \beta = 1.$$

In order to rescale this to the domain with $L = 1$ we must set $\beta = 1/2\pi$. Additionally, we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda = 1$. Therefore, for our scaled solution $\boldsymbol{v}$ to be a solution of the Euler equations we must set $\gamma = V_0 = 1/2\pi$.

I Rescaling the Euler-Voigt equations

Given a solution $\boldsymbol{u}_\alpha$ of the Euler-Voigt equations (1.4), we construct a new scaled solution withe construct a new scaled solution with

$$\tau = \lambda t, \qquad \kappa = \beta\alpha, \qquad \boldsymbol{y} = \beta\boldsymbol{x}.$$

where $\lambda, \beta > 0$ are constants. The variables $\boldsymbol{x}$ and α are both scaled by β because they are both spatial and we want to keep the relative size of the filter to the domain size fixed. Our new solution $\boldsymbol{v}$ is defined as

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{y}, \tau) = \gamma\boldsymbol{u}_\alpha(\boldsymbol{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t \boldsymbol{u}_\alpha = \frac{1}{\gamma}\frac{\partial \boldsymbol{v}}{d\tau}\frac{d\tau}{dt} = \frac{\lambda}{\gamma}\partial_\tau \boldsymbol{v}.$$

and

$$\boldsymbol{\nabla}\boldsymbol{u}_\alpha = \frac{1}{\gamma}\boldsymbol{\nabla}\boldsymbol{v}\frac{d\boldsymbol{y}}{d\boldsymbol{x}} = \frac{\beta}{\gamma}\boldsymbol{\nabla}\boldsymbol{v}.$$

and

$$\partial_t \boldsymbol{\Delta}\boldsymbol{u}_\alpha = \partial_t \frac{\beta^2}{\gamma}\boldsymbol{\Delta}\boldsymbol{v} = \frac{\lambda\beta^2}{\gamma}\partial_\tau \boldsymbol{\Delta}\boldsymbol{v}.$$

Substituting $\boldsymbol{v}$ into equation (1.4a)

$$-\alpha^2\frac{\lambda\beta^2}{\gamma}\partial_\tau\boldsymbol{\Delta v}+\frac{\lambda}{\gamma}\partial_\tau\boldsymbol{v}+\frac{\beta}{\gamma^2}(\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v}+\boldsymbol{\nabla}q=0,$$

where q is defined as whatever pressure is required to satisfy the incompressibility condition. This can be rearranged to the form,

$$-\kappa^2\partial_t\boldsymbol{\Delta v}+\partial_\tau\boldsymbol{v}+\frac{\beta}{\lambda\gamma}(\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v}+\boldsymbol{\nabla}q'=0.$$

Therefore, our function $\boldsymbol{v}$ can only be a solution to the Euler-Voigt equations if

$$\frac{\beta}{\lambda\gamma}=1.$$

For the Taylor Green Vortex with $L=2\pi$ and $V_0=1$, let $\gamma=\lambda=\beta=1$. In order to rescale this to the domain with $L=1$ we must set $\beta=1/2\pi$. Additionally, we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda=1$. Therefore, for our scaled solution $\boldsymbol{v}$ to be a solution of the Euler-Voigt equations we must set $\gamma=V_0=1/2\pi$.

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